12 Real-world example: An Akka betting house

This chapter covers

- Creating an elaborate use case
- Creating the entities and ports for them

This chapter presents a real-world example. The functionality is only part of a large business use case: sports betting. The idea is to create three actors at the business's core:

- The *market* is a sporting event with specific odds that can be bet against. For example, the Liverpool vs. Real Madrid soccer match has a market where bets can be placed on a home win, an away win, or a draw.
- The *wallet* represents a user's account with their balance.
- The *bet* is the amount of money invested in a market and secured by a wallet.

These entities reside in a cluster, so they can communicate directly. To communicate with them from the outside, the application has ports: two gRPC services and one Akka HTTP service. Figure 12.1 shows the big picture of how everything is related.